Most Common Kernel Functions

Linear Kernel

$$\boldsymbol{K}(a, b) = a^T b \qquad \text{No Mapping , } \phi(x) = x$$

Quadratic Kernel

$$\boldsymbol{K}(a, b) = [a^T b]^2 \text{ or } \qquad \boldsymbol{K}(a, b) = [a^T b + 1]^2$$

Polynomial Kernel (of degree “d”)

$$\boldsymbol{K}(a, b) = [a^T b]^d \text{ or } \qquad \boldsymbol{K}(a, b) = [a^T b + 1]^d$$

Radial Basis Function (RBF) Kernel

$$\boldsymbol{K}(a, b) = [e^{-\gamma \|a - b\|^2}]$$

γ is called RBF Kernel bandwidth